NNN-T-792a
March 29, 1963

SUPERSEDING
Int. Fed. Spec. NNN-T-00792 (ARMY-CmlC)
June 30, 1961 and
Fed. Spec. DD-T-721a
July 3, 1953

FEDERAL SPECIFICATION

TUBE, DRYING, LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass drying tubes for laboratory application.

1.2 Classification.

1.2.1 *Types and sizes.* Drying tubes, covered by this specification, shall be of the following types and sizes, as specified (see 6.2).

Type I—Straight, 1-bulb.

Size 1—16-26 by 100 millimeters (mm).
Size 2—17-31 by 150 mm.
Size 3—19-33 by 200 mm.

Type II—U-tube, Schwartz type.

Size 1—16 by 100 mm.
Size 2—18 by 150 mm.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

DD-G-541—Glass (Laboratory).
UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-up, Paperboard.
PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 626—Glassware; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside

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the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Uniform Classification Committee Publication:
Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

Commercial Standard:

CS21—Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Application for copies should be addressed to Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material. Drying tubes shall be of borosilicate glass in accordance with type I, class A of DD-G-541.

3.2 Design.

3.2.1 *Style.*

3.2.1.1 *Type I.* Drying tubes shall be straight and have one bulb. Each drying tube shall consist of two pieces of glass tubing, one larger than the other, united in approximately the same vertical axis by means of an expanded bulb of glass. The free end of the larger tube shall be tapered to fit a rubber stopper. The free end of the smaller tube shall be serrated or have a bulbous enlargement to provide for a secure fit for rubber tubing.

3.2.1.2 *Type II.* Drying tubes shall be U-shaped, the open end of each main arm terminating in a taper-ground joint. Each joint shall be fitted with a hollow taper-ground stopper. Perpendicular to and in the same plane as the two main arms of the U-tube, a side arm shall extend from each taper-ground joint. The free end of the side arm shall have a bulbous enlargement or be serrated to provide for a secure fit for rubber tubing. Each stopper shall have a side hole to coincide with the side arm opening in the joint and have an open bottom. Taper ground joints and stoppers shall be in accordance with Commercial Standard CS21.

TABLE I. *Dimensions of type I drying tubes*

Drying tube size	Dimensions					Rubber stopper accommodated
	Outside diameter			Length		
	Tube, mm.	Bulb, mm.	Stem, mm.	Tube*, mm.	Stem, mm.	
1	16 ± 1.0	26 ± 2.0	7-1/8 ± 1-1/8	100 ± 3.0	25 ± 3.0	0
2	17 ± 1.0	31 ± 2.0	7-1/8 ± 1-1/8	150 ± 4.0	25 ± 3.0	0
3	19 ± 1.0	33 ± 2.0	7-1/8 ± 1-1/8	200 ± 5.0	25 ± 3.0	1

* From top of tube to bottom of bulb.

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TABLE II. *Dimensions of type II drying tubes*

Drying tube size	Stopper, standard taper, size	Dimensions				
		Main arms			Side Arms	
		Outside diameter, mm.	Length*, mm.	Inside distance between arms, mm.	Outside diameter, mm.	Length, mm.
1	12/18 or 14/20	16 ± 1.0	100 ± 3.0	30 ± 5.0	7 ± 1.0	25 ± 3.0
2	14/20	18 ± 1.0	150 ± 4.0	38 ± 5.0	7 ± 1.0	25 ± 3.0

* From top of main arms to center line of bend.

3.2.2 *Dimensions.* Dimensions of drying tubes shall be as shown in table I and table II, respectively, for type I and type II drying tubes.

3.3 **Annealing.** Ring strain shall be no greater than that of the standard reference disc No. 1 when tested as specified in 4.6.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.4 **Identification marking.** Each drying tube shall be permanently and legibly marked with the manufacturer's name or trademark.

3.5 **Workmanship.** Drying tubes shall be symmetrical in shape and free of cracks and other defects that impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.2.1.2, and section 5 have been met.

4.3 **Lotting.** A lot shall consist of those drying tubes of the same type and size produced by one manufacturer for delivery at one time.

4.4 **Sampling.**

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* A sample of 2 drying tubes shall be taken at random from each 100 drying tubes or fraction thereof in the lot.

4.5 **Examination.** Samples of drying tubes and level A and level B preparation for delivery thereof shall be examined in

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accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Drying tubes.*

Categories	Defects
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Dimensions not as specified
102	Stopper and joint do not fit properly or not as specified
103	Shape not as specified (symmetrical)
104	Defective (cracks or other defects)
105	Marking incomplete, missing, or illegible

4.5.1.2 *Level A and B preparation for delivery (section 5).*

Categories	Defects
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Unit, intermediate, or shipping container not as specified
102	Number of tubes per intermediate container not as indicated
103	Unlike tubes packaged per intermediate or shipping container
104	Closure of unit, intermediate, or shipping container not as specified
105	Unit, intermediate, or shipping container damaged
106	Cushioning material or fiberboard fittings not as specified
107	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete

4.6 **Tests.**

4.6.1 *Annealing.* Annealing shall be tested in accordance with method No. 420 of Fed. Test Method Std. No. 626.

4.7 **Acceptance/rejection criteria.** If any drying tube fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each drying tube shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636, type I, class 1; or PPP-B-676, type, class, and style optional. Drying tubes, provided with glass stoppers, shall have the stopper secured in the box. The box shall be provided with fiberboard fittings or sufficient cushioning material conforming to type II, class B of PPP-C-843 to prevent shifting of contents. Box closure shall be secured with tape conforming to UU-T-111.

5.1.1.2 *Intermediate packaging.* The number of drying tubes, of the same type and size, as specified (see 6.2) shall be packaged in a carton of appropriate size conforming to PPP-B-636, type I, class 1, style RSC. The box shall be closed in accordance with the appendix of the box specification.

5.1.2 *Level C.* Drying tubes shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Drying tubes shall be packed in a box conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Con-

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tainers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 *Level B.* Drying tubes shall be packed in fiberboard boxes conforming to PPP-B-636, type I or II, class 1, style RSC, minimum bursting strength 275 pounds per square inch.

5.2.3 *Level C.* Drying tubes shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Drying tubes covered by this specification are intended for use with calcium chloride or other laboratory chemical drying agents.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and size required.

(*c*) Level of packaging and packing required.

(*d*) Number of tubes per intermediate and shipping container required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Transportation description.** Transportation description applicable to this item is:

Glassware, chemical laboratory.
Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Glassware, laboratory.
Motor volume minimum weight 18,000 pounds.

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